

# Predicting Hit Songs Using Repeated Chorus

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## 1. Problem and motivation

The goal is to measure whether short chorus audio contains useful information about stronger chart success. Class 1 is a song in the source year-end Billboard Hot 100 collection; class 0 is another song sampled from weekly Hot 100 charts. Both classes can have charted. The available labels are retrospective proxies, so this is not a prospective hit forecast or a claim that chorus sound causes popularity.

## 2. Data and representation

The inherited dataset has 751 songs from 71 normalized artist-name groups: 366 year-end hits and 385 other chart songs. Each row contains 518 summary features from 15-second chorus audio, including MFCC, chroma, RMS energy, tonnetz, spectral and zero-crossing measurements. The latest representation selects 74 standard-deviation columns that summarize within-chorus variation. Artist and title are identification/grouping metadata only.

The fixed original split gives 597 development songs across 52 groups and 154 historical songs across 19 other groups. Matching original recordings, complete version identities and fully verified chart labels are not available. The historical partition was already examined in earlier work and was not reused to select V4.

## 3. Training and evaluation

The V4 experiment retained 55 V3 controls and added eight predeclared balanced logistic-regression settings using 74 standard-deviation features, two scalers and four regularization strengths. Five outer artist-grouped folds estimate the model-selection procedure; three inner artist-grouped folds choose settings. Imputation, variance filtering, scaling and any feature selection fit on training rows only. Native classifier thresholds were kept fixed.

The selected final development model is logistic regression ( $C=0.1$ , balanced classes) with standard scaling and 74 features. The app accepts the original ordered 518-feature input and applies the saved selection inside the model pipeline.

## 4. Results on the same development songs

| Metric                   | Previous V3 | Current V4 |
|--------------------------|-------------|------------|
| Accuracy                 | 53.27%      | 54.61%     |
| Balanced accuracy        | 53.25%      | 54.66%     |
| Precision (year-end hit) | 51.86%      | 53.07%     |
| Recall (year-end hit)    | 52.76%      | 56.55%     |
| F1 score                 | 52.31%      | 54.76%     |

The paired change in balanced accuracy is +1.41 points; approximate 95% interval -4.29 to +7.33 points. V4 balanced accuracy interval: 50.72%-58.50%. These intervals resample artist groups and condition on saved predictions. V3 and V4 reuse the same 597 previously inspected songs, so this is exploratory evidence, not an independent new test.

# Implementation, limits and conclusion

## 5. What the software does

The Python project validates the ordered feature table, preserves artist-group split membership and records model settings, predictions, fold assignments and checksums. A Streamlit application provides historical-song examples, exploratory audio upload and a results view. The original baseline remains the default; V4 can be selected as a research option.

The repository includes a README, pinned dependencies, data provenance, the working app, saved models and run manifests. Individual fit records are stored in compact trial archives; their original hashes remain in each manifest and are checked when a run is loaded. The latest local test suite passed 30 tests. The current presentation file is retained for later work.

## 6. What the result means

V4 increased ordinary accuracy from 53.27% to 54.61% and balanced accuracy from 53.25% to 54.66% on matched grouped development folds. It changed the final selected classifier to a 74-feature setting. The paired uncertainty interval includes zero, so these data do not establish a reliable gain. None of the five reported V4 metrics reaches 60%, the team's next exploratory goal.

The 60% goal is not an assignment requirement. The available PDF guidelines assess quality of deliverables, working code, repository, write-up, presentation, live demonstration, Q&A; and individual contribution; they do not state a numeric accuracy cutoff.

## 7. Limitations

Artist-name grouping cannot fully resolve aliases and credited collaborators. Labels and recording versions have incomplete independent verification. Original waveforms are unavailable, so live upload extraction cannot be proved equivalent to the inherited training features. Repeated use of this development set limits the strength of new improvement claims. The old 154-song set is a historical benchmark; a genuinely new locked test set is still missing.

## 8. Conclusion and next work

The project has a functioning, reproducible ML pipeline and an honest measurement of the current representation. Chorus variation shows a small exploratory increase, but the task remains difficult. To pursue 60%, the strongest next step is verified song/artist/chart evidence, permitted matching recordings and a reserved new evaluation collection before another model search.

## References and reproducibility

[1] Feature collection and source license. [2] Eric Liu, original repeated-chorus study. [3] Project code and saved evidence. Local run: v4\_std74\_001; source configuration: configs/v4\_std74.json.